

Day 28: AI & Advanced Analytics in Power BI

📌 Definition to Remember

Power BI integrates native Machine Learning and Generative AI capabilities to automate insight discovery, time-series forecasting, automated anomaly detection, and natural language querying (**Q&A & Copilot**).

1. Built-in Machine Learning Visuals

1. Key Influencers Visual:

- Uses logistic/linear regression to rank what factors drive a specific business outcome (e.g., "What factors influence a customer churn status being True?").

2. Decomposition Tree with AI Splits:

- Automatically finds the dimension with the highest or lowest contribution to a metric drilldown.

3. Smart Narrative:

- Auto-generates dynamic text summaries of active charts that recalculate when slicers change.

2. Time-Series Forecasting & Anomaly Detection

- **Forecasting:** Adds predictive confidence bands (70%, 80%, 95% confidence intervals) to line charts based on exponential smoothing algorithms.
- **Anomaly Detection:** Identifies statistically significant outliers in time-series data with automated diagnostic explainers.

3. Q&A & Copilot Integration

- **Q&A Natural Language Visual:** Allows non-technical stakeholders to type plain-English questions (e.g., "Show top 5 sales reps by profit in 2025 as a bar chart").
- **Copilot in Power BI:** AI assistant capable of drafting DAX measures, summarizing report pages, and suggesting chart layouts.

★ Must-Master Skills & Interview Points

1. Configuring the Key Influencers visual for churn, lead conversion, or CSAT scoring.
2. Tuning time-series forecasting seasonality and confidence intervals.
3. Optimizing the Semantic Model Q&A linguistic schema with synonyms.

⚡ Quick Recall

Key Influencers (Regression drivers) + Anomaly Detection (Outliers) + Smart Narrative + Q&A / Copilot